

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Civil Engineering) SEMESTER - 5 Summer 2025 ( Supply. )**

**Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 5**

**Subject Code & Name: BTCVC501 - DESIGN OF STEEL STRUCTURES**

**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

**Q1. Objective type questions. (Compulsory Question)**

**12**

- 1 What is the hole diameter for a bolt of 25 mm diameter ?
  - A) 25 mm
  - B) 26 mm
  - C) 30 mm
  - D) 28 mm
- Which force is NOT typically acting on a gantry girder?
  - A) Vertical wheel load
  - B) Wind load
  - C) Impact load from crane
  - D) Earthquake uplift
- 3 What is the purpose of lacing in built-up columns?
  - A) Increase flexural rigidity
  - B) Distribute axial load evenly
  - C) Prevent buckling of individual components
  - D) Connect base plates
- 4 Plastic hinge formation occurs at:
  - A) Maximum axial point
  - B) Point of zero bending moment
  - C) Point of maximum bending moment
  - D) Mid-span always
- 5 What is a typical factor of safety used in steel design?
  - A) 1.0
  - B) 1.5
  - C) 2.5
  - D) 3.0

- 6 Failure in bolted connections due to shear generally occurs along:  
 A) Centreline of bolt  
 B) Outer fiber of plate  
 C) Net section  
 D) Bolt shank
- 7 The deflection of steel beams in buildings other than industrial building is limited to span divided by  
 A) 180  
 B) 250  
 C) 300  
 D) 325
- Collapse mechanism in plastic analysis involves formation of:  
 A) Cracks  
 B) Hinges  
 C) Nodes  
 D) Bolts
- 9 In IS 800-2007, the design strength of a beam is calculated considering:  
 A) Allowable stress  
 B) Yield stress  $\times$  section modulus  
 C) Moment capacity with partial safety factor  
 D) Section area only
- The design wind speed is  $V$ . The design wind pressure will be given as  
 A)  $0.4 V^2$   
 B)  $0.5 V^2$   
 C)  $0.6 V^2$   
 D)  $0.8 V^2$
- 11 The number of plastic hinges required for collapse in a simply supported beam is:  
 A) 1  
 B) 2  
 C) 3  
 D) Infinite
- 12 Which of the following is a disadvantage of steel structures?  
 A) Uniformity of material  
 B) High strength-to-weight ratio  
 C) Susceptibility to corrosion  
 D) Ease of fabrication

Q2. Give the following.

- A) List and explain different types of standard rolled steel sections used in steel structures. 6
- B) What are the different methods of steel design? Compare Working Stress Method and Limit State Method. 6

Q3. Solve the following.

- A) Determine the design bending strength of ISLB 350@486 N/m considering the beam to be laterally unsupported. The design shear force  $V$  is less than the design shear strength. The unsupported length of the beam is 3.0 m. 6

- B) Design a double-angle section to carry a tension of 300 kN. The end connection is to be made using M20 bolt of class 4.6. Assume that the angles (2 ISA 100x75x8mm @ 7.5 kg/m) are provided on both sides of the gusset plate. 6
- Q4. Solve Any Two of the following.
- A) Calculate the maximum wheel load on gantry girder for the following data: 6
- Crane capacity = 100 kN
  - Self weight of crane girder excluding trolley = 150 kN
  - Self weight of trolley electric motor, hook etc = 35 kN
  - Approximate minimum approach of the crane = 1.0 m
  - Wheel base = 2.5 m
  - c/c distance between gantry rails = 20 m
  - c/c distance between column (span of gantry girder) = 7.5 m
  - Self weight of rail section = 300 N/m
  - Yield stress of steel = 250 N/mm<sup>2</sup>
- B) Design an I section purlin, for an industrial building situated at Allahabad, to support a galvanised corrugated iron sheet roof for the following data : 6
- Spacing of truss c/c = 6.0 m
  - Span of truss = 12 m
  - Slope of truss = 30°
  - Spacing of purlin = 1.5 m
  - Intensity of wind pressure = 2 kN/m<sup>2</sup>
  - Weight of galvanised sheets = 130 N/m<sup>2</sup>
  - Grade of steel = Fe 410
- C) Discuss the various structural components of an industrial shed. What are the functional roles of truss members, purlins, and supports in the overall structural system? 6
- Q5. Solve Any Two of the following.
- A) Design a slab base for a column ISHB 350 @ 67.4 kg/m carrying a factored load of 1000 kN. Also design the welded connection between slab base and column. Use concrete grade M20 and steel of Fe 410. 6
- B) A column section consisting of 2 ISMC 250 which are placed face to face at spacing of 400 mm between webs. Determine the capacity of section if the height of column is 6m having both ends of column held in position but not restrained against rotation. Also design suitable bracing system. 6
- What are column splices? Why and where are they provided? Briefly explain their design considerations. 6
- Q6. Solve Any Two of the following.
- A) Define plastic hinge. Explain its role in determining the collapse load of a steel structure. Support your answer with neat sketches. 6
- B) A simply supported beam of span L of circular section is subjected to central point load W find 6
- Shape factor of circular section
  - Length of plastic hinge
- C) What is a collapse mechanism in plastic analysis? Explain the concept in detail and describe any two types of collapse mechanisms with neat sketches. 6